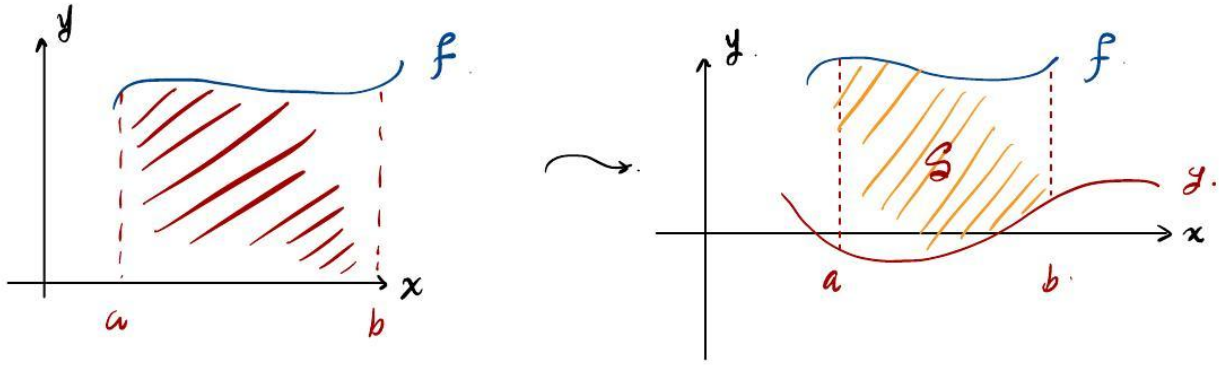
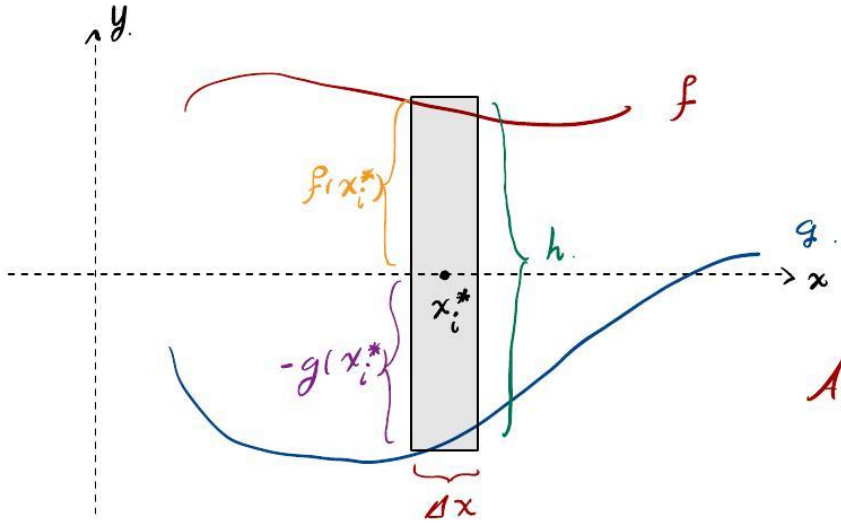
Lecture 11, July 27.

Math 225 - Summer

## 5.1. Areas between Curves.




Consider the region $S$ that lies between two curves $y=f(x)$ and $y=g(x)$ and the vertical lines $x=a$ and $x=b$. were $f$ and $g$ are continuous functions and $f(x) \geq g(x)$.



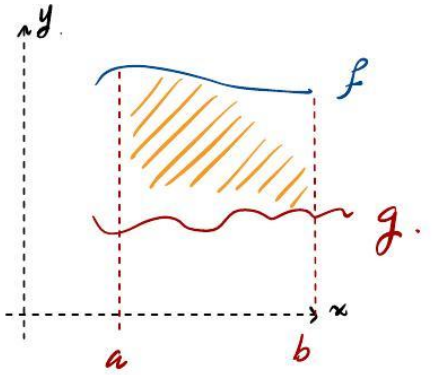

$$A_R = \Delta x \cdot h = \Delta x \{ f(x_i^*) - g(x_i^*) \}.$$

The approximation appears to become better and better as $n \to \infty$ (where $n$ is the number of rectangles). Therefore, we define the area $A$ of the region $S$ as

$$A = \lim_{n \to \infty} \sum_{i=1}^{n} [f(x_i^*) - g(x_i^*)] \Delta x.$$

**Def.** The area $A$ of the region bounded by the curves $y=f(x)$ and $y=g(x)$ and lines $x=a$, $x=b$ where $f, g$ are continuous and $f(x) \geq g(x)$ for all $x \in [a,b]$ is

$$A = \int_a^b [f(x) - g(x)] dx.$$




In the case where $f, g \geq 0$ then

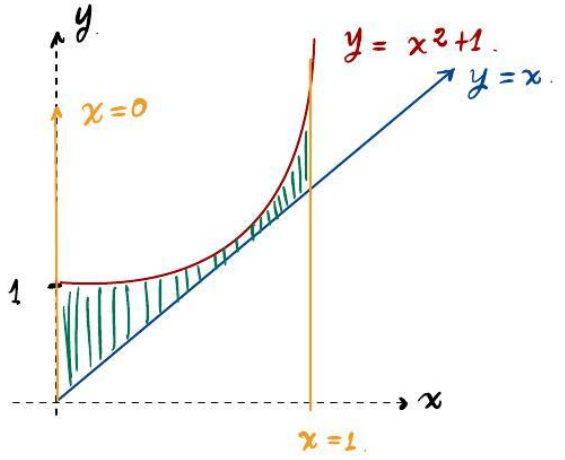
$A = [\text{area under } y=f(x)] - [\text{area under } y=g(x)]$.

$$= \int_a^b f(x)\,dx - \int_a^b g(x)\,dx.$$

$$= \int_a^b [f(x) - g(x)]\,dx$$

**Example.** Find the area of the region bounded above by $y = x^2+1$. bounded below $y=x$, and bounded on the sides by $x=0$ and $x=1$.

**Sln:**




limits are $0 \leq x \leq 1$.

$$A = \int_0^1 [(x^2+1) - (x)]\,dx.$$

$$= \int_0^1 (x^2 - x + 1)\,dx.$$

$$= \frac{x^3}{3} - \frac{x^2}{2} + x \Big|_0^1 = \frac{1}{3} - \frac{1}{2} + 1 = \frac{5}{6}.$$

**Example.** Find the area of the region enclosed by the parabolas $y = x^2$ and $y = 2x - x^2$.

**Sln:** We first find the points of intersection of the parabolas by solving their equation simultaneously.

$$x^2 = 2x - x^2.$$

$$2x^2 - 2x = 0$$

$$x(x-1) = 0 \quad \Rightarrow \quad x = 0, \quad x = 1.$$

Let $x = \frac{1}{2}$ $\quad f(x) = x^2 \quad \Rightarrow f(\frac{1}{2}) = \frac{1}{4}$

$$g(x) = 2x - x^2 \Rightarrow g(\tfrac{1}{2}) = 1 - \frac{1}{4} = \frac{3}{4}.$$

Since $g(\frac{1}{2}) \geq f(\frac{1}{2})$ then $g(x) \geq f(x)$ for all $x \in [0,1]$.

$$A = \int_0^1 \{g(x) - f(x)\}\,dx = \int_0^1 \{(2x - x^2) - (x^2)\}\,dx = \int_0^1 (2x - 2x^2)\,dx$$

$$= x^2 - \frac{2}{3}x^3 \Big|_0^1 = 1 - \frac{2}{3} = \frac{1}{3}.$$

**Example.** Find the approximate area of the region bounded by the curves

$$y = \frac{x}{\sqrt{x^2+1}}, \quad y = x^4 - x.$$

**Sln:** We can't solve the equation $x^4 - x = \frac{x}{\sqrt{x^2+1}}$ we can approximate the integration limits using geogebra.

$$A = \int_0^{1.18} \left\{ \frac{x}{\sqrt{x^2+1}} - (x^4 - x) \right\} dx = \int_0^{1.18} \frac{x}{\sqrt{x^2+1}}\, dx - \int_0^{1.18} (x^4 - x)\, dx.$$

$\int \frac{x}{\sqrt{x^2+1}}\, dx$ Let $u = x^2+1$ $du = 2x\,dx \Rightarrow x\,dx = \frac{du}{2}$

by the substitution rule

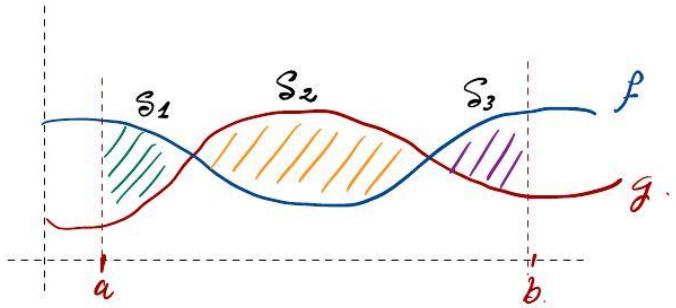
$$\int \frac{x}{\sqrt{x^2+1}}\, dx = \frac{1}{2}\int \frac{1}{\sqrt{u}}\, du = \frac{1}{2}\int u^{-1/2}\, du = \sqrt{u} = \sqrt{x^2+1}$$

then,

$$\int_0^{1.18} \frac{x}{\sqrt{x^2+1}}\, dx = \sqrt{x^2+1} \Big|_0^{1.18} = \sqrt{(1.18)^2+1} - 1.$$

$$A = \sqrt{(1.18)^2+1} - 1 - \left( \frac{x^5}{5} - \frac{x^2}{2} \Big|_0^{1.18} \right) = \sqrt{2.39} - 1 - \left( \frac{(1.18)^5}{5} - \frac{(1.18)^2}{2} \right)$$

If we are asked to find the area between $y = f(x)$ and $y = g(x)$ where $f(x) \geq g(x)$ for some values of $x$ but $g(x) \geq f(x)$ for other values of $x$.



then we split the region $S$ into several regions $S_1, S_2, S_3, \ldots$ with areas $A_1, A_2, A_3, \ldots$

We then define the area of the region $S$ to be the sum of the areas of the smaller regions. $A = A_1 + A_2 + A_3$.

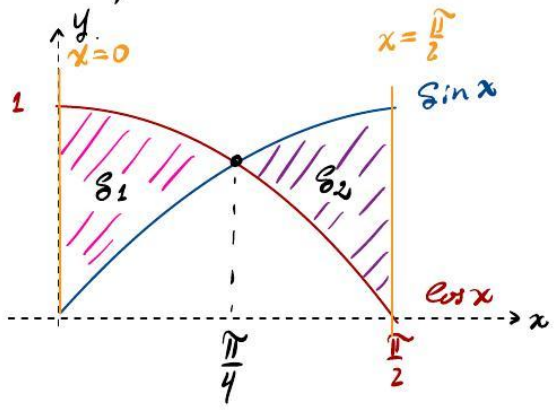
$$|f(x) - g(x)| = \begin{cases} f(x) - g(x) & \text{if } f(x) \geq g(x) \\ g(x) - f(x) & \text{if } g(x) \geq f(x). \end{cases}$$

**Def.** The area between the curves $y = f(x)$ and $y = g(x)$ and $x = a$, $x = b$ is

$$A = \int_a^b |f(x) - g(x)|\, dx.$$

**Example.** Find the area of the region bounded by the curves $y = \sin x$, and $y = \cos x$, $x = 0$, $x = \frac{\pi}{2}$.

**Sln:** The point of intersection occur when $x = \frac{\pi}{4}$.




$$A = \int_0^{\frac{\pi}{2}} |\cos x - \sin x|\, dx = A_1 + A_2.$$

$$= \int_0^{\frac{\pi}{4}} (\cos x - \sin x)\, dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x - \cos x)\, dx$$

($A_1$ under the first integral, $A_2$ under the second)

$$= (\sin x + \cos x)\Big|_0^{\frac{\pi}{4}} + (-\cos x - \sin x)\Big|_{\frac{\pi}{4}}^{\pi/2}$$

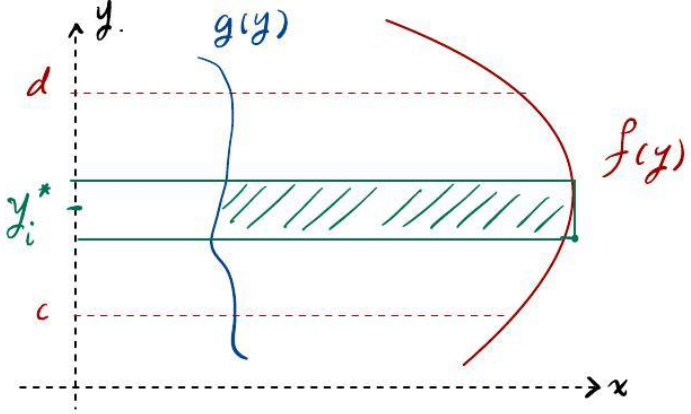
$$= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - (0+1)\right) + \left(-0 - 1 - \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right)\right)$$

$$= 2\sqrt{2} - 2.$$

**Areas between curves: (Integrating with respect $y$).**

Some regions are best treated by regarding $x$ as a function of $y$.

If the region bounded by the curves $x = f(y)$ $x = g(y)$ $y = c$ and $y = d$ where $f, g$ are continuous functions and $f(y) \geq g(y)$ for $y \in [c, d]$ then




$$A = \int_c^d [f(y) - g(y)]\,dy.$$

**Example.** Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

**Sln.** By solving $\begin{cases} x = y + 1 = f(y) \\ x = \dfrac{y^2 - 6}{2} = g(y). \end{cases}$

$$y + 1 = \frac{y^2 - 6}{2} \iff 2y + 2 = y^2 - 6$$

$$\iff y^2 - 2y - 8 = 0$$

$$\iff (y - 4)(y + 2) = 0$$

$$\iff y = 4, \quad y = -2$$

If $y = 0$ then $f(0) = 1$, $g(0) = -3$ then $f(y) \geq g(y)$ for $-2 \leq y \leq 4$.

$$A = \int_{-2}^{4} [f(y) - g(y)]\,dy = \int_{-2}^{4} \left[y + 1 - \left(\frac{y^2}{2} - 3\right)\right] dy$$

$$= \int_{-2}^{4} \left(-\frac{y^2}{2} + y + 4\right) dy$$

$$= -\frac{1}{6}y^3 + \frac{y^2}{2} + 4y \Big|_{-2}^{4}$$

$$= -\frac{1}{6}(4^3) + \frac{4^2}{2} + 4^2 + \frac{1}{6}(2)^3 - 2 - 8 = 18.$$

**Problem for you:** Find the area of the region enclosed by $x + 2y = 3$ $y = x$ and $y = \frac{1}{4}x$ using.

a). $x$ as variable of integration

b). $y$ as variable of integration.